CLASS 10 · CBSE · SAMPLE PAPER

MathSetu Sample Paper 1: Mathematics (Standard)

Time allowed: 3 hours · Maximum marks: 80

GENERAL INSTRUCTIONS

1. This paper has 38 questions in five sections, A to E. All questions are compulsory.
2. Section A has 20 questions of 1 mark each: 18 multiple choice questions and 2 assertion and reason questions.
3. Section B has 5 questions of 2 marks each.
4. Section C has 6 questions of 3 marks each.
5. Section D has 4 questions of 5 marks each.
6. Section E has 3 case study questions of 4 marks each, with parts of 1, 1 and 2 marks.
7. There is no overall choice. An internal choice is given in 2 questions of Section B, 2 questions of Section C, 3 questions of Section D and in the 2-mark part of each question of Section E.
8. Draw neat figures wherever they are needed. Take $\pi = \frac{22}{7}$ unless another value is given.
9. Calculators are not allowed.

1 Section A (20 questions of 1 mark each)

1. The HCF of $2^3 \times 3^2 \times 5$ and $2^2 \times 3^3 \times 7$ is: (a) 12 (b) 36 (c) 72 (d) 108 1 mark

2. If α and β are the zeroes of $x^2 - 7x + 10$, then $\frac{1}{\alpha} + \frac{1}{\beta}$ equals: (a) $\frac{7}{10}$ (b) $\frac{10}{7}$ (c) $-\frac{7}{10}$ (d) $\frac{3}{10}$ 1 mark

3. A quadratic polynomial whose zeroes are 3 and -4 is: (a) $x^2 - x - 12$ (b) $x^2 + x + 12$ (c) $x^2 + x - 12$ (d) $x^2 - 7x + 12$ 1 mark

4. The pair of equations $3x + ky = 5$ and $6x + 8y = 7$ has no solution when k equals: (a) 2 (b) 4 (c) 8 (d) 16 1 mark

5. The roots of $2x^2 - 5x + 3 = 0$ are: (a) real and equal (b) real and distinct (c) not real (d) both negative 1 mark

6. What is the 15th term of the AP $8, 14, 20, \ldots$? (a) 86 (b) 92 (c) 98 (d) 104 1 mark

7. Find the sum of the first 10 terms of the AP $3, 7, 11, \ldots$: (a) 190 (b) 200 (c) 210 (d) 220 1 mark

8. The distance between the points $(-3, 4)$ and $(5, -2)$ is: (a) 6 units (b) 8 units (c) 10 units (d) 14 units 1 mark

9. The mid-point of the line segment joining $(2, -5)$ and $(-6, 9)$ is: (a) $(-2, 2)$ (b) $(-4, 4)$ (c) $(4, -7)$ (d) $(2, -2)$ 1 mark

10. In $\triangle ABC$, D lies on AB and E lies on AC with $DE \parallel BC$. If $AD = 3$ cm, $DB = 4.5$ cm and $AE = 4$ cm, then EC is: (a) 5 cm (b) 5.5 cm (c) 6 cm (d) 7.5 cm 1 mark

11. Two similar triangles have corresponding sides in the ratio $3 : 5$. If the perimeter of the smaller triangle is 27 cm, the perimeter of the larger triangle is: (a) 36 cm (b) 40 cm (c) 45 cm (d) 75 cm 1 mark

12. In $\triangle ABC \sim \triangle PQR$, $\angle A = 50^\circ$ and $\angle Q = 60^\circ$. Then $\angle C$ is: (a) 50° (b) 60° (c) 70° (d) 80° 1 mark

13. A point is 13 cm from the centre of a circle of radius 5 cm. The length of the tangent from the point to the circle is: (a) 8 cm (b) 12 cm (c) 14 cm (d) 18 cm 1 mark

14. PA and PB are tangents to a circle with centre O. If $\angle APB = 70^\circ$, then $\angle AOB$ is: (a) 70° (b) 90° (c) 110° (d) 140° 1 mark

15. If $\sin\theta = \dfrac{3}{5}$ and θ is acute, then $\tan\theta$ is: (a) $\dfrac{4}{5}$ (b) $\dfrac{3}{4}$ (c) $\dfrac{4}{3}$ (d) $\dfrac{5}{3}$ 1 mark

16. The value of $2\tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$ is: (a) 0 (b) 1 (c) 2 (d) 3 1 mark

17. The area of a sector of a circle of radius 6 cm with central angle 60° is: (a) 3π cm^2 (b) 6π cm^2 (c) 9π cm^2 (d) 12π cm^2 1 mark

18. For a distribution, the mean is 20 and the median is 22. Using the empirical relation, the mode is: (a) 18 (b) 24 (c) 26 (d) 28 1 mark

19. **Assertion (A):** When a die is thrown once, the probability of getting a prime number is $\dfrac{1}{2}$. **Reason (R):** The prime numbers on a die are 2, 3 and 5. Choose the correct option: (a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A. (c) A is true, but R is false. (d) A is false, but R is true. 1 mark

20. **Assertion (A):** Two cubes, each of edge 4 cm, are joined end to end. The surface area of the cuboid formed is 192 cm^2. **Reason (R):** When two solids are joined, the surface area of the new solid is less than the sum of their surface areas. Choose the correct option from the four options given in Question 19. 1 mark

2 Section B (5 questions of 2 marks each)

21. Two alarm bells ring at intervals of 18 minutes and 24 minutes. They ring together at 8:00 a.m. At what time will they next ring together? 2 marks

22. Find the value of k for which $kx^2 - 12x + 9 = 0$ has two equal roots. **OR** Solve for x and y: $2x + 3y = 13$ and $5x - 2y = 4$. 2 marks

23. In $\triangle ABC$, points D and E lie on AB and AC with $AD = 4$ cm, $DB = 6$ cm, $AE = 5$ cm and $EC = 7.5$ cm. Show that $DE \parallel BC$. If $BC = 15$ cm, find DE. 2 marks

24. Evaluate $4\sin^2 60^\circ + 3\tan^2 30^\circ - 8\sin 45^\circ \cos 45^\circ$. **OR** If $\sin(A + B) = 1$ and $\cos(A - B) = \dfrac{\sqrt{3}}{2}$, where $0^\circ < A + B \le 90^\circ$ and $A > B$, find A and B. 2 marks

25. A bag has 5 red, 8 white and 7 green balls. One ball is drawn at random. Find the probability that the ball is (i) white (ii) not red. 2 marks

3 Section C (6 questions of 3 marks each)

26. Prove that $\sqrt{5}$ is an irrational number. 3 marks

27. Find the zeroes of $6x^2 - 7x - 3$ and verify the relationship between the zeroes and the coefficients. **OR** The sum of the digits of a two-digit number is 9. The number obtained by reversing the digits is 27 more than the original number. Find the number. 3 marks

28. Show that the two tangent segments drawn to a circle from a point outside it are equal in length. 3 marks

29. Prove that $\dfrac{\sin\theta}{1+\cos\theta} + \dfrac{1+\cos\theta}{\sin\theta} = 2\operatorname{cosec}\theta$. 3 marks

30. In a circle of radius 14 cm, a chord subtends a right angle at the centre. Find the area of the minor segment it cuts off. **OR** The minute hand of a clock is 10.5 cm long. Find the area it sweeps in 20 minutes. 3 marks

31. Find the mean of the following distribution of marks. 3 marks

Marks	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Number of students	3	5	12	6	4

4 Section D (4 questions of 5 marks each)

32. A bus covers 300 km at a uniform speed. If its speed had been 10 km/h more, the journey would have taken 1 hour less. Find the speed of the bus. **OR** In an AP, the 4th term is 11 and the sum of the first 10 terms is 155. Find the AP, its 20th term and the sum of its first 20 terms. 5 marks

33. Prove that if a line is drawn parallel to one side of a triangle to cut the other two sides in distinct points, then it divides those two sides in the same ratio. Use this result to find x if, in $\triangle PQR$, $ST \parallel QR$ with S on PQ and T on PR, and $PS = x$, $SQ = x - 2$, $PT = x + 2$, $TR = x - 1$. 5 marks

34. From the top of a tower, the angles of depression of the top and the bottom of a pole 10 m high are 30° and 60°. Find the height of the tower and its distance from the pole. **OR** Two ships are on opposite sides of a lighthouse 75 m high, in line with its foot. From the top of the lighthouse their angles of depression are 30° and 45°. Find the distance between the ships. (Take $\sqrt{3} = 1.732$.) 5 marks

35. A solid toy is a cone standing on a hemisphere of the same radius, 7 cm. The total height of the toy is 31 cm. Find its volume and its total surface area. **OR** A solid metal cone of radius 6 cm and height 24 cm is melted and recast into solid spheres of radius 2 cm. Find the number of spheres, and the surface area of one sphere. (Take $\pi = 3.14$.) 5 marks

5 Section E (3 case study questions of 4 marks each)

36. A student starts a fitness plan. She does 12 push-ups on the first day and, on every following day, 3 more than on the day before. (i) How many push-ups does she do on the 10th day? (ii) On which day does she do 60 push-ups? (iii) Find the total number of push-ups in the first 15 days, **OR** find the number of days in which the total reaches 255. 4 marks, case study

37. Three lamp posts in a park stand at the points $P(2, 1)$, $Q(8, 3)$ and $R(6, 9)$ on a grid whose unit is 1 metre. (i) Find the distance PQ. (ii) Find the mid-point of QR. (iii) Find the point that divides PR in the ratio $1 : 3$, **OR** show that $\triangle PQR$ is an isosceles right triangle. 4 marks, case study

38. A housing society recorded the water used in one day by each of its 40 households, as shown in the table. (i) Write the modal class. (ii) How many households used less than 200 litres? (iii) Find the median of the data, **OR** find its mode. 4 marks, case study

Water used (litres)	100 – 150	150 – 200	200 – 250	250 – 300	300 – 350
Number of households	5	8	13	9	5

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